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§ 1.8 Gaussian Elimination with Pivoting.

Example Solve $Ax = b$, where

$$A = \begin{bmatrix} 0 & 4 & 1 \\ 1 & 1 & 3 \\ 2 & -2 & 1 \end{bmatrix}$$

$$A \xrightarrow{P_{12}} \begin{bmatrix} 1 & 1 & 3 \\ 0 & 4 & 1 \\ 2 & -2 & 1 \end{bmatrix} \xrightarrow{R_3 - 2R_1} \begin{bmatrix} 1 & 1 & 3 \\ 0 & 4 & 1 \\ 0 & -4 & -5 \end{bmatrix}$$

$$\xrightarrow{R_3 + R_2} \begin{bmatrix} 1 & 1 & 3 \\ 0 & 4 & 1 \\ 0 & 0 & -4 \end{bmatrix} = U \text{ (upper triangular)}$$

Really

$$\underbrace{\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{P_{12}} A = \begin{bmatrix} 1 & 1 & 3 \\ 0 & 4 & 1 \\ 2 & -2 & 1 \end{bmatrix}$$

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}}_{L_1} \begin{bmatrix} 1 & 1 & 3 \\ 0 & 4 & 1 \\ 2 & -2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 3 \\ 0 & 4 & 1 \\ 0 & -4 & -5 \end{bmatrix}$$

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}}_{L_2} \begin{bmatrix} 1 & 1 & 3 \\ 0 & 4 & 1 \\ 0 & -4 & 5 \end{bmatrix} = U$$

$$\Longleftrightarrow \quad L_2 L_1 P A = U$$

$$\Longleftrightarrow \quad PA = L_1^{-1} L_2^{-1} U = \underset{\text{lower}}{L}\ \underset{\text{upper}}{U}$$

$$\Leftrightarrow \quad A = P^{-1}LU, \quad P^{-1} = P^T$$

permutation matrices are unitary matrices

$$= P^T L U \qquad P^T P = I$$

<u>Fact</u> If permutations $P_1, P_2, \ldots, P_k$ are needed during Gaussian Elimination, then

$$P = P_1 P_2 \cdots P_k$$

can be applied to A in advance and then <u>no</u> permutations are need to produce $PA = LU$

<u>Notes</u>

(1) There is a strategy to move the largest in magnitude entry in a column to the pivotal position (thru permutations) in order to control the growth of entries in the process. This process is called <u>partial pivoting</u>

(E.g. in example above we would have used P_{13} instead of P_{12})

In fact there's something even better than partial pivoting: <u>Complete pivoting</u>

$$\underset{n\times n}{A} \to \cdots \begin{bmatrix} \blacksquare & \cdots & \cdots \\ 0 & & \\ \vdots & & \\ 0 & & \end{bmatrix}$$

to be reduced $(n-1)\times(n-1)$

largest in modulus entry in trailing $(n-1)\times(n-1)$

Use permutations of rows & columns to put $\circledast$ in the pivotal position

<u>Example</u> $A = \begin{bmatrix} 0 & 1 & 3 \\ 2 & 3 & -4 \\ 1 & 1 & 1 \end{bmatrix}$ Move $(2,3)$ entry into $(1,1)$

max modulus

$$\xrightarrow[Q_{13}]{P_{12}} \begin{bmatrix} -4 & 3 & 2 \\ 3 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} = P_{12} A Q_{13}$$

rows

column

In the end we obtain

$$PAQ = LU, \quad P, Q \text{ permutations}$$

<u>Theorem</u> For every invertible $A \in M_n(\mathbb{R})$ there exist permutation matrices P, Q & triangular matrices L (lower) & U (upper) such that $PAQ = LU$.

So $A = P^T L U Q^T$

$(Ax = b \Rightarrow x = Q U^{-1} L^{-1} P b)$

(2) Pivoting minimizes the "growth" of modulus of the entries. (i.e: "Growth factor", ≤ 2 for diagonally dominant matrix)

(3) FLOP count:

Without pivoting:

$$2n^2 + 2(n-1)^2 + \dots = \sum_{k=1}^{n} k^2 = \frac{2}{3} n^3$$

$$O(n^3)$$

With pivoting: Additional

$(n-k)$ comparisons at k-th step

$$\sum_{k=1}^{n-1} (n-k) \approx \frac{n^2}{2}$$

$$O(n^3 + n^2) = O(n^3)$$

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Back to § 1.4

§ 1.4 Positive Definite Systems; Cholesky Decomposition

A special kind of matrix arising in applications:

Concepts & definitions

- $A \in M_n(\mathbb{R})$
- A symmetric : $A = A^T$ (transpose) $(= A^*$ hermitian$)$

$$A = [a_{ij}], \quad A^T = [a_{ji}]$$

- If A above satisfies

$$x^T A x > 0, \quad \forall x \neq 0, \ x \in \mathbb{R}^n$$

then we call A <u>positive definite</u>

<u>Example</u> Let M be any invertible matrix in $M_n(\mathbb{R})$. Form

$$A = M^T M \ (\text{or } MM^T)$$

Then $A^T = (M^T M)^T = M^T (M^T)^T = M^T M = A$

& $\forall x \neq 0$

$$x^T A x = x^T M^T M x = (Mx)^T (Mx) = y^T y$$

where $y = Mx \neq 0$ since M is invertible, & $x \neq 0$

and $y^T y = \|y\|_2^2 > 0$.

<u>Corollary</u> A positive definite $\Rightarrow$ A invertible

$\Rightarrow$ $Ax = b$ has unique solution.

<u>Theorem</u> (Cholesky Decomposition Theorem)

Let $A \in M_n(\mathbb{R})$ be positive definite.

Then A can be uniquely decomposed into the product

$$A = R^T R, \quad R = [r_{ij}]$$

where R is upper triangular with $r_{ii} > 0$ for $i = 1, 2, \ldots, n$

<u>Proof</u>: Wait / we need:

<u>Proposition 1</u> If $A = [a_{ij}]$ is positive definite, then $a_{ii} > 0$, $i = 1, 2, \ldots, n$

<u>Proof</u> Let e_i be i-th column of $n \times n$ I. Then

$$a_{ii} = e_i^T A e_i > 0, \quad i = 1, 2, \ldots, n \quad \blacksquare$$

Proposition 2

If $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \in M_n(\mathbb{R})$

is a block partition of a positive definite matrix, where A_{11}, A_{22} are square ($k \times k$, $(n-k) \times (n-k)$, resp.), then A_{11}, A_{22} are <u>also</u> positive definite

<u>Proof</u> Let $x_1 \in \mathbb{R}^k$, $x_1 \neq 0$, $A_{11} \in M_k(\mathbb{R})$.

Then

$$x_1^T A_{11} x_1 = [x_1^T \; 0^T] A \begin{bmatrix} x_1 \\ 0 \end{bmatrix} > 0$$

$\in \mathbb{R}^n \setminus \{0\}$

Of course $A_{11}^T = A_{11}$. Thus A_{11} is positive definite. Similarly, A_{22} is positive definite. ∎

<u>Note</u> Positive definiteness is preserved (prove!) under permutational similarity

PAP^T, $P^T = P^{-1}$ = permutation matrix

Thus <u>every</u> principal submatrix of a pos. defin. matrix is also positive definite.

<u>Proposition 3</u> If $A, X \in M_n(\mathbb{R})$, where A is positive definite & X is invertible, then

$$B = X^T A X \quad \text{(congruence)}$$

is also positive definite.

<u>Proof</u> Let $y \in \mathbb{R}^n$, $y \neq 0$. Then

$$y^T B y = y^T (X^T A X) y = (\underbrace{Xy}_{\neq 0})^T A (Xy) > 0$$

$$\& \quad B^T = B \quad \blacksquare$$

<u>Back to proving Cholesky Decomp.</u>

By induction on n:

$\boxed{n=1}$ $A = [a_{11}]$, A pos. def $\Rightarrow a_{11} > 0$

Thus $A = (\sqrt{a_{11}})^T \cdot \sqrt{a_{11}} = R^T R$

$R = [\sqrt{a_{11}}]$ upper triangular.

$\boxed{\text{Hypothesis}}$ Every $(n-1) \times (n-1)$ pos. def. admits the unique Chol. decomposition.

Now let

$$A = \begin{bmatrix} a_{11} & b^T \\ b & \hat{A} \end{bmatrix} \in M_n(\mathbb{R})$$

be positive definite,

$$\hat{A} \in M_{n-1}(\mathbb{R}), \quad b \in \mathbb{R}^{n-1}$$

let $r_{11} = \sqrt{a_{11}} > 0 \quad (a_{11} > 0)$

$$s = r_{11}^{-1} b \in \mathbb{R}^{n-1}$$

$$\tilde{A} = \hat{A} - ss^T$$

Then we can write:

$$A \underset{\text{verify}}{=} \begin{bmatrix} r_{11} & 0^T \\ s & I_{n-1} \end{bmatrix} \begin{bmatrix} 1 & 0^T \\ 0 & \tilde{A} \end{bmatrix} \begin{bmatrix} r_{11} & s^T \\ 0 & I_{n-1} \end{bmatrix}$$

Let $\quad X = \begin{bmatrix} r_{11} & s^T \\ 0 & I_{n-1} \end{bmatrix}^{-1}$. Then X

is invertible. Consider

$$B = X^T A X$$

By Prop. 3, B is also pos. definite.
By Prop 2, $\hat{A}$ & $\tilde{A}$ are pos. definite.
So by inductive hypothesis

$$\tilde{A} = \tilde{R}^T \tilde{R},$$

where $\tilde{R}$ is $(n-1)\times(n-1)$ (unique) upper triangular with positive diag. entries

So

$$A = \begin{bmatrix} v_{11} & 0^T \\ s & I \end{bmatrix} \begin{bmatrix} 1 & 0^T \\ 0 & \tilde{R}^T \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & \tilde{R} \end{bmatrix} \begin{bmatrix} v_{11} & s^T \\ 0 & I \end{bmatrix}$$

$$= \begin{bmatrix} v_{11} & 0^T \\ s & \tilde{R}^T \end{bmatrix} \underbrace{\begin{bmatrix} v_{11} & s^T \\ 0 & \tilde{R} \end{bmatrix}}_{R}$$

$$= R^T R \quad \text{as desired.} \quad \blacksquare$$

<u>Notes</u>

(1) In view of this theorem & the example above we have

$$\boxed{\begin{array}{c} A \in M_n(\mathbb{R}) \text{ is posit. definite} \\ \text{iff} \\ A = M^T M, \quad M \in M_n(\mathbb{R}) \text{ is invertible.} \end{array}}$$

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(Notes continued)

(2) In the proof of Cholesky decomp. a useful concept is hidden in the factorization:

Schur Complement

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \in M_n(\mathbb{R})$$

$A_{11} \in M_k(\mathbb{R})$ & invertible.

Assume also that we can proceed with Gaussian Elimination of A without pivoting.

So the first step in the process is to put zeros under the first pivot. Proceed to the second pivot, third, ..., k^{th} pivot.

What does the outcome look like as a matrix.

$$A \to \cdots \to \begin{bmatrix} \boxed{\diagdown^{*}_{0}} & * \\ 0 & B \end{bmatrix}$$

$B \in M_{n-k}(\mathbb{R})$ looks like

$$B = A_{22} - A_{21} A_{11}^{-1} A_{12}$$

This B is called the "<u>Schur Complement</u> of A_{11} in A". Denoted by

$$B = A/A_{11}$$

Interesting properties: e.g.

$$\det(A/A_{11}) = \frac{\det A}{\det A_{11}}$$

Iterative Methods for solving linear systems

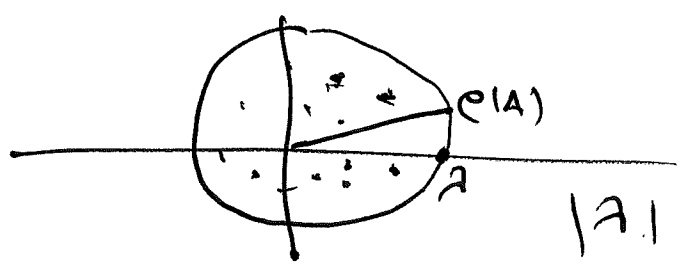
- Faster that direct method
- Accurate
- Preconditioning can be used to accelerate & improve accuracy.

Some preliminaries

(1) Let $A \in M_n(\mathbb{R})$. The spectral radius of A is

$$\rho(A) = \max\{|\lambda| : \lambda \text{ eigenvalue of } A\}$$

rho



$|\lambda| = \rho(A)$

We call A convergent if and only if $\rho(A) < 1$. It means that

$$\lim_{m \to \infty} A^m = 0 \quad \text{(zero matrix)}$$

<u>Theorem</u> $A \in M_n(\mathbb{R})$ is convergent iff $\rho(A) < 1$

<u>Proof</u> (when A is diagonalizable; more complicated proof using Jordan Canonical Form done in §11).

A diagonalizable $\rightsquigarrow$ $\exists$ invertible $S \in M_n(\mathbb{C})$ such that

$$S^{-1} A S = D = \begin{bmatrix} \lambda_1 & & & 0 \\ & \lambda_2 & & \\ & & \ddots & \\ 0 & & & \lambda_n \end{bmatrix}$$

$\lambda_1, \dots, \lambda_n$ eigenvalues of A.

Then $A^m = (SDS^{-1})^m = \dots = SD^m S^{-1}$

So $\lim_{m \to \infty} A^m = S \lim_{m \to \infty} D^m S^{-1} = 0$ iff

$|\lambda_j|^m \to 0$, $m \to \infty$. That is, $|\lambda_j| < 1$

$j = 1, 2, \dots, n$. ∎

(2) $A \in M_n(\mathbb{R})$, $A = [a_{ij}]$ is called <u>diagonally dominant</u> if

$$|a_{ii}| \geq \sum_{\substack{j=1 \\ j \neq i}}^{n} |a_{ij}|, \quad i = 1, 2, \ldots, n.$$

"Strictly" if $>$ is used in all rows.

(3) <u>Gershgorin's Theorem</u>
(Gersgorin)

Let $A \in M_n(\mathbb{R})$, $A = [a_{ij}]$.
All the eigenvalues of A lie in the set

$$G = \bigcup_{i=1}^{n} R_i, \text{ where}$$

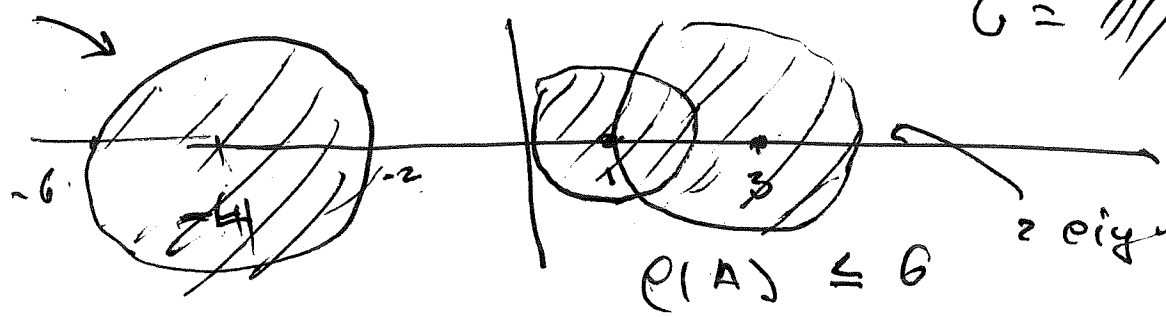
$$R_i = \left\{ z \in \mathbb{C} : |z - a_{ii}| \leq \sum_{\substack{j=1 \\ j \neq i}}^{n} |a_{ij}| \right\}$$

E.g.

$$A = \begin{bmatrix} -4 & 1 & -1 \\ 0 & 1 & 1 \\ 1 & -1 & 3 \end{bmatrix}$$

1 eigenvalue

$G =$ ////



2 eigenvalues

$\rho(A) \leq 6$

So $\frac{A}{6.1}$ is convergent?

(because $\rho(cA) = |c|\rho(A) < 1$ if $|c| < \frac{1}{\rho(A)}$.)

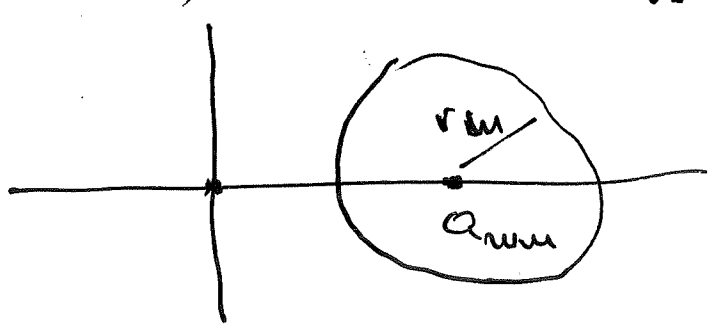
Application of Gersgorin

Theorem (Levy-Desplanques)

Let $A \in M_n(\mathbb{R})$ be strictly diagonally dominant. Then A is invertible.

Proof All Gersgorin discs of A are centered at the diagonal entries which by strict diagonal dominance are farther away ~~that~~ from 0 than the radius of each disc (sum of moduli of off-diag. entries)

$r_m < |a_{mm}|$



So $0 \notin$ Gersgorin region, so 0 is not eigenvalue ∎

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Iterative Methods

(Out of
- Nonnegative Matrices in the Mathematical Sciences by Berman & Plemmons
- Iterative Matrix Analysis by Richard Varga.)

Solve $Ax = b$, $A \in M_n(\mathbb{R})$, $b \in \mathbb{R}^n$

Idea "Split A": $A = M - N$, $M, N \in M_n(\mathbb{R})$

Then $(M - N)x = b$

$$\Rightarrow \quad Mx = Nx + b$$

Assume (design) that M is invertible.

Thus
$$\boxed{x = M^{-1}Nx + M^{-1}b} \qquad (*)$$

- x appears on both sides of $(*)$
- $(*)$ "solves" for x

So $(*)$ leads itself to "iteration":

$$\boxed{x^{(k+1)} = M^{-1}N x^{(k)} + M^{-1}b}$$

$x^{(k)}$ is the k-th iterate

$k = 0, 1, 2, \ldots$ $x^{(0)}$ = initial

What can we say/hope about

$$\{x^{(0)}, x^{(1)}, \dots\}?$$

<u>Issue 1</u> When will $\{x^{(k)}\}$ converge as $k \to \infty$? Hopefully (we aim) that $x^{(k)} \longrightarrow x$ (solution to the system).

Define $\varepsilon^{(k)} = x^{(k)} - x$. If $\varepsilon^{(k)} \to 0$, then $x^{(k)} \longrightarrow x$ as $k \to \infty$.

But

$$\begin{aligned}
\varepsilon^{(k)} = x^{(k)} - x &= M^{-1}N x^{(k-1)} + M^{-1}b - x \\
&\overset{(*)}{=} M^{-1}N x^{(k-1)} + \cancel{M^{-1}b} - M^{-1}Nx - \cancel{M^{-1}b} \\
&= M^{-1}N(x^{(k-1)} - x) \\
&= M^{-1}N\,\varepsilon^{(k-1)} \\
&= (M^{-1}N)^2\,\varepsilon^{(k-2)} \\
&= \dots = (M^{-1}N)^k\,\varepsilon^{(0)}
\end{aligned}$$

For $\varepsilon^{(k)}$ to converge to 0 it is necessary & sufficient that

$$(M^{-1}N)^k \longrightarrow 0$$

(convergent matrix)

So we need:

$$\rho(M^{-1}N) < 1$$

So we have a way to construct a iterative method to solve a system.

<u>Issue 2</u> By design of our splitting

$$\varepsilon^{(k)} \to 0 \quad \text{as } k \to \infty.$$

So how do we find such splittings?

The choice of M, N is tailored to special classes of matrices, e.g. "diagonally dominant" (: happens when discretizing pde's)

Three classical methods:

<u>Jacobi Method</u>

$$A = D - E - F, \text{ where}$$

D = diagonal entries $a_{ii} \neq 0$ (invertible)

E = strictly upper triangular part of A $(j > i,\ a_{ij})$

F = strictly lower triang. part of A $(a_{ij},\ i > j)$

~~Note~~ For $A = M - N$, $M = D$, $N = E + F$

For convergence, we need

$$\rho\left(D^{-1}(E+F)\right) < 1$$

This occurs e.g. when A is strictly diagonally dominant, because

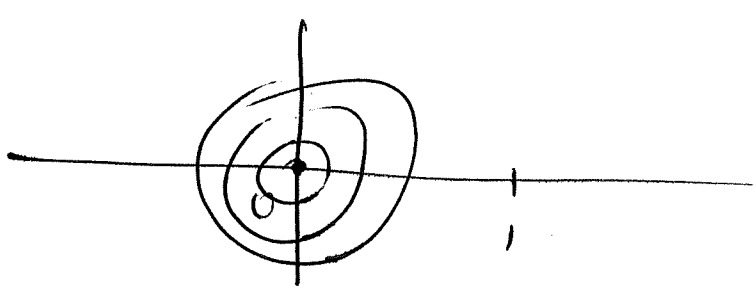
$$0 \neq |a_{ii}| > \sum_{\substack{j=1 \\ j\neq i}}^{n} |a_{ij}| \qquad (i=1,2,\dots,n)$$

$$\Rightarrow \sum_{j\neq i} \frac{|a_{ij}|}{|a_{ii}|} < 1 \qquad (**)$$

However,

$$\left[D^{-1}(E+F)\right]_{ij} = \begin{cases} \dfrac{a_{ij}}{a_{ii}}, & j\neq i \\[2mm] \dfrac{0}{a_{ii}} = 0, & j=i \end{cases}$$

So applying Gerschgorin's Theorem to $D^{-1}(E+F)$ we have that all Gerschgorin discs are centered at 0



and have radius < 1 (by $(**)$)

So $\rho\left(D^{-1}(E+F)\right) < 1$

Thus Jacobi method should work for diag. dominant matrices

<u>Note</u> Sometimes we can scale A in $Ax=b$ to make the coefficient strictly diagonally dominant & apply Jacobi:

Solve $\quad GAx = Gb$, G invertible

preconditioner
(e.g. diag. matrix so that GA is str. diag. dominant)

$(\Longleftrightarrow Ax = b)$

Gauss-Seidel

Splitting $A = L + U$

U = strictly upper triangular part

L = lower triangular part

(assume L is invertible)

<u>Example</u>

$$A = \begin{bmatrix} 2 & -1 & 1 \\ 1 & 3 & 0 \\ 0 & -1 & 2 \end{bmatrix} = \underbrace{\begin{bmatrix} 2 & 0 & 0 \\ 1 & 3 & 0 \\ 0 & -1 & 2 \end{bmatrix}}_{L} + \underbrace{\begin{bmatrix} 0 & -1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}}_{U}$$

$$(L+U)x = b \Rightarrow Lx = b - Ux$$

$$\Rightarrow Lx^{(k+1)} = b - Ux^{(k)}$$

$$\Rightarrow x^{(k+1)} = -L^{-1}Ux^{(k)} + L^{-1}b$$

Convergence iff

$$\rho(-L^{-1}U) = \rho(L^{-1}U) < 1$$

Have <u>two</u> nice classes work well
- strictly diag. dom.
- positive definite